Kavitha has 3 beads more than Kannan and Aravind have together. So, the number of beads that Kavitha has is given by $x+x+20+3$.

Now, we change the situation. Instead of Aravind has 20 more beads than Kannan, suppose Aravind has 20 beads less than Kannan.

If Kavitha has 3 beads more than what Kannan and Aravind have together, then find the number of beads that Kavitha has.

Now, Aravind has x – 20 beads. So, the number of beads that Kavitha has, is given by $x + x - 20 + 3$.

For example, to add $8ab$, $4ab$ and $2ab$, where ab is variable, we add the co-efficients alone, that is $8ab + 4ab + 2ab = (8 + 4 + 2)\ ab = 14\ ab = 14ab$.

In the same way, we can add the algebraic expressions. Let us try to add the expressions $11y + 7$ and $5y - 3$, where $11y$ and $5y$ are like terms with a variable y and 7 and -3 are constants (like terms).

$$\begin{aligned}\text{Hence, } (11y + 7) + (5y - 3) &= [11y + 5y] + [7 + (-3)] \\ &= (11 + 5)\ y + (7 - 3) \\ &= 16y + 4.\end{aligned}$$

Try this

Add the terms: i) $3p$, $14p$ ii) m, $12m$, $21m$ iii) $11abc$, $5abc$
iv) $12y$, $-y$ v) $4x$, $2x$, $-7x$

Example 3.5 Add the expressions: (i) $pq - 1$ and $3pq + 2$ (ii) $8x + 3$ and $1 - 7x$

Solution

$$\begin{aligned}\text{i) } (pq - 1) + (3pq + 2) &= (pq + 3pq) + (-1 + 2) \\ &= (1 + 3)pq + 1 \\ &= 4pq + 1 \\ \text{ii) } (8x + 3) + (1 - 7x) &= 8x + 3 + 1 - 7x \\ &= (8x - 7x) + (3 + 1) \\ &= (8 - 7)\ x + 4 \\ &= x + 4.\end{aligned}$$

Subtraction of a term can be looked as addition of its additive inverse as we saw in subtraction of integers. To understand subtraction of algebraic expression, let us consider subtraction of algebraic expression with one term.

For example, to subtract $6y$ from $12y$, we can add $12y$ and $(-6y)$.

Thus we have, $12y + (-6y) = 12y - 6y$

$= (12 - 6)y = 6y$.

Now, let us subtract $-mn$ from $3mn$. Additive inverse of $-mn$ is mn. Hence, we have to add mn with $3mn$. Thus, $3mn + mn = (3 + 1)mn = 4mn$.

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